

WEEK 3

Recap+while/for loops



Recap



Input/print/ variables

```
age = int(input("How old are you? "))  
next_year_age = age + 1
```

```
city = input("What city do you live in? ")  
temperature = float(input("What is the current temperature in °C? "))  
# Using an f-string to embed variables directly  
print(f"It is currently {temperature}°C in {city}.")
```

```
price = float(input("Enter item price: ")) total_price = price * 1.08  
# Adding 8% tax  
print("Total price with tax:", total_price)
```

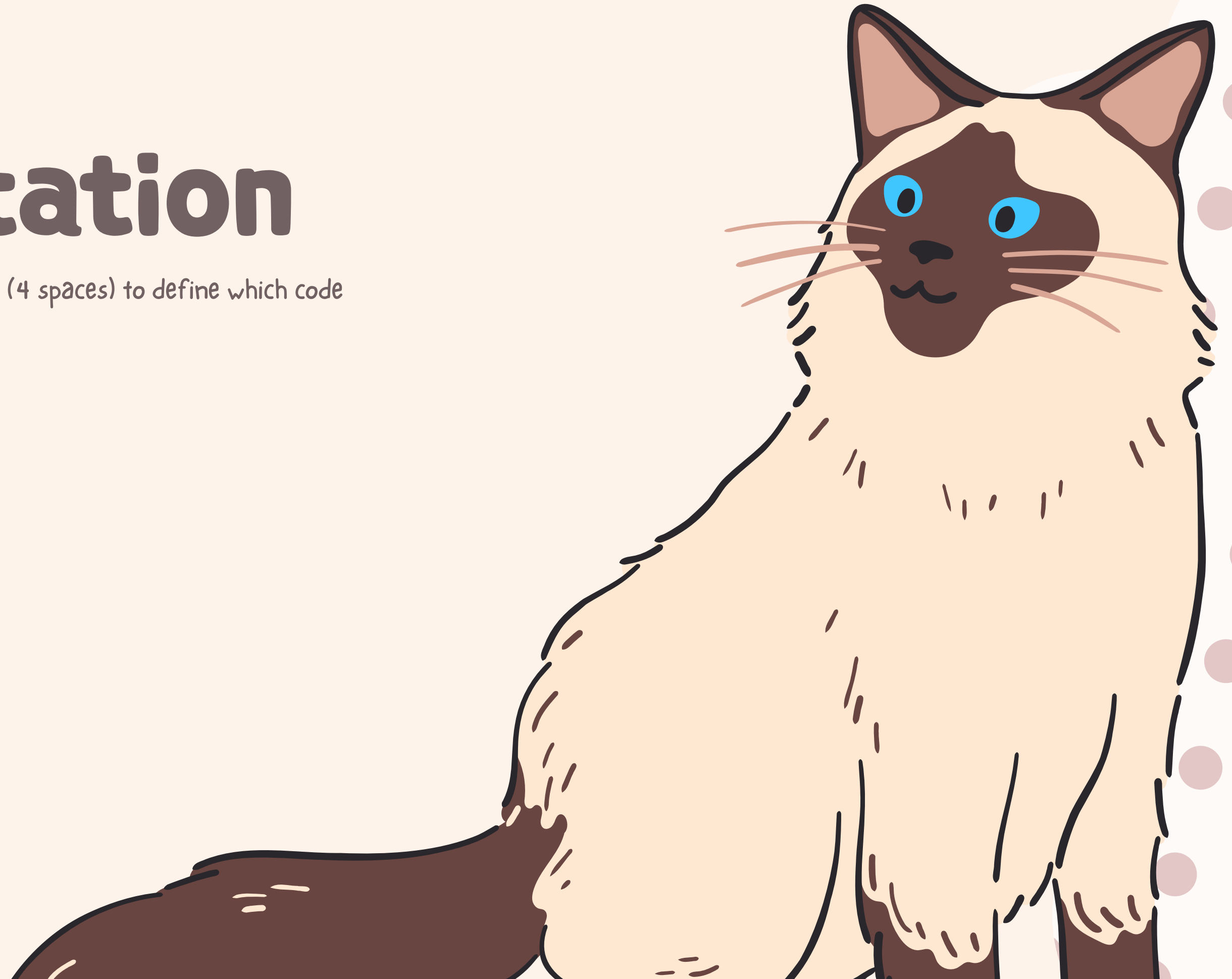


If/else statements

```
temperature = float(input("Enter the temperature in °C: "))  
if temperature > 25:  
    print("It's a warm day, dress light!")  
else:  
    print("It's cool outside, grab a jacket.")
```

Indentation

Note: Python uses indentation (4 spaces) to define which code belongs inside the if block.





**Little more
examples**

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What python code is this?

```
score = int(input("Enter your test score (0-100): "))
```

```
if score >= 90:  
    print("Grade: A")  
elif score >= 80:  
    print("Grade: B")  
elif score >= 70:  
    print("Grade: C")  
else:  
    print("Grade: Needs Improvement")
```



All of the previous ones!

```
score = int(input("Enter your test score (0-100): "))
if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
elif score >= 70:
    print("Grade: C")
else:
    print("Grade: Needs Improvement")
```

For and While loops



For loop

```
# Print a message 5 times (counts from 0 to 4)
for i in range(5):
    print(f"Loop iteration number: {i}")
```

How range() works:

- range(5) generates numbers 0, 1, 2, 3, 4 (starts at 0 by default, stops before 5).
- range(1, 6) generates numbers 1, 2, 3, 4, 5 (start, stop).
- range(2, 11, 2) generates 2, 4, 6, 8, 10 (start, stop, step size).

For loop

```
fruits = ["apple", "banana", "cherry"]
```

```
# Process each item in the list one by one  
for fruit in fruits:  
    print(f"I like {fruit}s!")
```



For loop

Combining for Loops with if / else

```
numbers = [12, 5, 8, 21, 3, 14]
```

```
for num in numbers:
```

```
    if num % 2 == 0:
```

```
        print(f"{num} is Even")
```

```
    else:
```

```
        print(f"{num} is Odd")
```

break and continue

- break: Stops the loop completely when a condition is met.
- continue: Skips the rest of the current iteration and moves immediately to the next item.

Example of break: Search for an item and stop early

```
items = ["bread", "milk", "eggs", "apples"]
```

```
for item in items:
```

```
    if item == "eggs":
```

```
        print("Found eggs! Stopping search.")
```

```
        break
```

```
    print(f"Checked {item}...")
```



While loop

- for loop: Used when you know in advance how many times to repeat something (e.g., "repeat 5 times" or "for every item in a list").
- while loop: Used when you want to keep repeating code as long as a condition remains true (e.g., "keep asking until the user types 'quit'").

Learn More →

While loop

Always make sure the condition eventually becomes False (like updating count). Otherwise, you will create an infinite loop that runs forever!

```
count = 1
```

```
# Runs as long as count is less than or  
equal to 5
```

```
while count <= 5:
```

```
    print(f"Current count: {count}")
```

```
    count += 1 # Increment count by 1  
(same as count = count + 1)
```

```
print("Loop finished!")
```

while True:

**secret_code = input("Enter the secret code to
pass: ")**

if secret_code == "python123":

print("Access granted!")

break # Immediately exits the while loop

print("Wrong code! Try again.\n")

THANK YOU

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Github:

<https://github.com/Mariam-eng211/YoungMakers-Introductory-Python-Curriculum-for-Emerging-Programmers>